第 9 章 重积分

二重积分计算及应用

习题课

主要内容

举　　例

课堂练习

主要内容

定义 $$\iint_D f(x,y)d\sigma = \lim_{\lambda\to 0}\sum_{i=1}^{n} f(x_i,y_i)\Delta\sigma_i$$

(与分法无关、点的取法无关）

性质　与定积分相类似的性质（线性性、*对称性*、对区域的可加性、比较性、估值、中值）

★计算

选择坐标 $\begin{cases}\text{利用直角坐标}\\ \text{利用极坐标} \\ \text{利用变量代换}\end{cases} \Rightarrow$ 化二次积分

对称性

1. D关于x轴对称（x轴上方部分为D_1)

$$\text{则}\iint_D f(x,y)d\sigma=\begin{cases}2\iint_{D_1} f(x,y)d\sigma & \text{当} f(x,y)=f(x,-y)\\ 0 & \text{当} f(x,-y)=-f(x,y)\end{cases}$$

2. D关于y轴对称（y轴右边部分为D_1)

$$\text{则}\iint_D f(x,y)d\sigma=\begin{cases}2\iint_{D_1} f(x,y)d\sigma & \text{当} f(x,y)=f(-x,y)\\ 0 & \text{当} f(-x,y)=-f(x,y)\end{cases}$$

3. D关于x轴、y轴均对称（第一象限部分为D_1)

则 $$\iint_D f(x,y)d\sigma = \begin{cases} 4\iint_{D_1} f(x,y)d\sigma & \text{当}f(x,-y = f(x,y) = f(-x,y) \\ 0 & \text{当}f(-x,y) = -f(x,y) \text{或}f(x,-y = -f(x,y) \end{cases}$$

应用

几何应用

1.求平面区域D的面积： $\sigma = \iint_D d\sigma$

2.求曲顶柱体体积：

$$V = \iint_D f(x,y)d\sigma$$ （顶： $f(x,y)>0$）

3.求曲面S的面积：

设 $z = f(x,y),(x,y) \in D$

$$A = \iint_D \sqrt{1 + f_x{}^2 + f_y{}^2}\, d\sigma$$

物理应用

1.求非均匀分布平面薄片D的质量

$$M=\iint_D \rho(x,y)d\sigma$$

2.求非均匀分布平面薄片D的重心

$$\bar{x}=\frac{\iint_D x\rho(x,y)d\sigma}{M},\ \bar{y}=\frac{\iint_D y\rho(x,y)d\sigma}{M}$$

3.求非均匀分布平面薄片D绕坐标轴及的转动惯量

4.平面薄片对质点的引力

举　例

例1 设连续函数$f(x,y)$满足

$$f(x,y)=|x|e^{y}-x\cos(xy)+\iint_D f(x,y)dxdy$$

其中D是由曲线$y=|x|$及$y=x^2$围成的平面有界区域，求$f(x,y)$。

解 $$\iint_D f(x,y)dxdy=a \Rightarrow f(x,y)=|x|e^{y}-x\cos(xy)+a$$

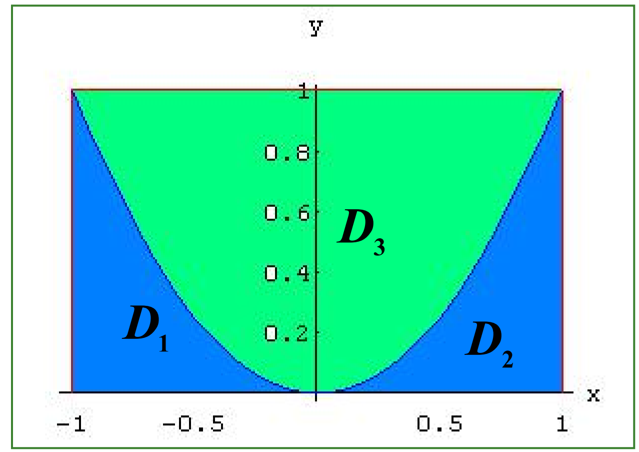
$$a=\iint_D f(x,y)dxdy=2\int_0^1 dx\int_{x^2}^{x} xe^{y}dy+a\iint_D dxdy$$

$$=3-e+\frac{a}{3} \Rightarrow a=\frac{9-3e}{2}$$

$$\therefore f(x,y)=|x|e^{y}-x\cos(xy)+\frac{9-3e}{2}$$

例2 计算 $\iint\limits_{D}|y-x^{2}|d\sigma$. 其中 $D:-1\le x\le 1, 0\le y\le 1$.

解 先去掉绝对值符号，如图



$$\iint\limits_{D}|y-x^{2}|d\sigma$$

$$=\iint\limits_{D_1+D_2}(x^{2}-y)d\sigma+\iint\limits_{D_3}(y-x^{2})d\sigma$$

$$=\int_{-1}^{1}dx\int_{0}^{x^2}(x^{2}-y)dy+\int_{-1}^{1}dx\int_{x^2}^{1}(y-x^{2})dy=\frac{11}{15}.$$

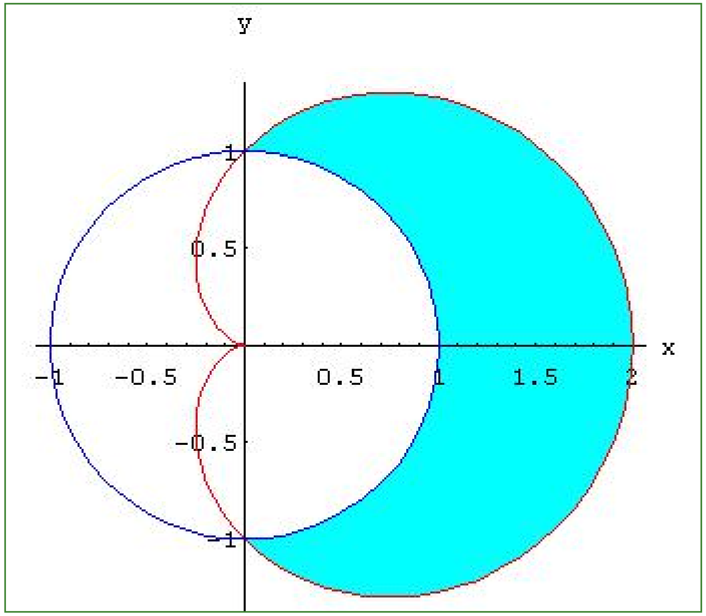
例4 计算 $\iint_D \sqrt{x^2+y^2}\,d\sigma$. 其中 D 是由心脏线 $r=a(1+\cos\theta)$ 和圆 $r=a$ 所围的面积（取圆外部）.

解 $\iint_D \sqrt{x^2+y^2}\,d\sigma$

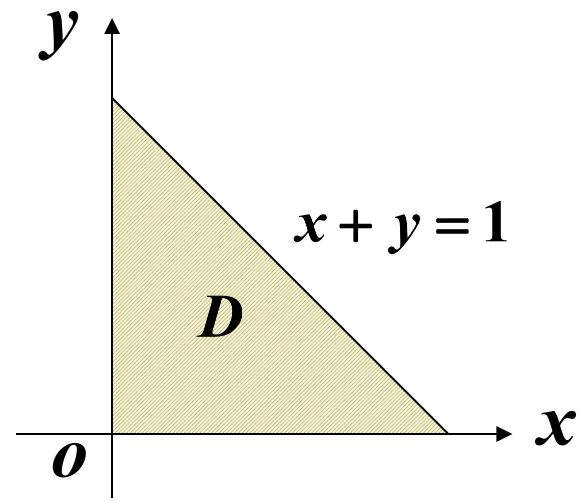
$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_a^{a(1+\cos\theta)} r\cdot r\,dr$$

$$=\frac{1}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a^3[(1+\cos\theta)^3-1]\,d\theta$$

$$=a^3\left(\frac{22}{9}+\frac{\pi}{2}\right).$$



例4 计算 $I=\iint\limits_D \cos\left(\dfrac{x-y}{x+y}\right)dxdy$. 其中 D 由 $x+y=1$, $x=0$ 及 $y=0$ 所围成.



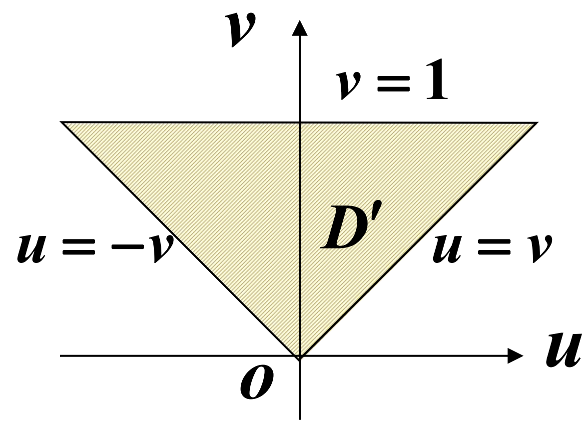
解 令 $u=x-y, v=x+y$,

则 $x=\dfrac{u+v}{2}, y=\dfrac{v-u}{2}$.

$D\to D'$, 即 $x=0\to u=-v$;

$y=0\to u=v$;

$x+y=1\to v=1$.



$$J=\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}\frac{1}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{1}{2}\end{vmatrix}=\frac{1}{2},$$

故 $$I=\iint_{D'}\cos\frac{u}{v}|J|dudv$$

$$=\frac{1}{2}\int_0^1 dv\int_{-v}^{v}\cos\frac{u}{v}du$$

$$=\frac{1}{2}\int_0^1 2\sin 1\cdot v dv=\frac{1}{2}\sin 1.$$

例5 求曲面$z=2-x^2-y^2$

与$z=x^2+y^2$所围成的体积．

解 积分区域D的边界是这两曲面交线

在xoy坐标面上的投影．

消去方程组：$\begin{cases} z=2-x^2-y^2 \\ z=x^2+y^2 \end{cases}$中的$z$得$\begin{cases} x^2+y^2=1 \\ z=0 \end{cases}$

$$V=\iint\limits_D (2-x^2-y^2)d\sigma-\iint\limits_D (x^2+y^2)d\sigma$$

$$=2\iint\limits_D (1-x^2-y^2)d\sigma=4\cdot 2\int_0^1 dx\int_0^{\sqrt{1-x^2}}(1-x^2-y^2)dy$$

$$=8\int_0^1 \frac{2}{3}(1-x^2)^{\frac{3}{2}}dx=\pi$$

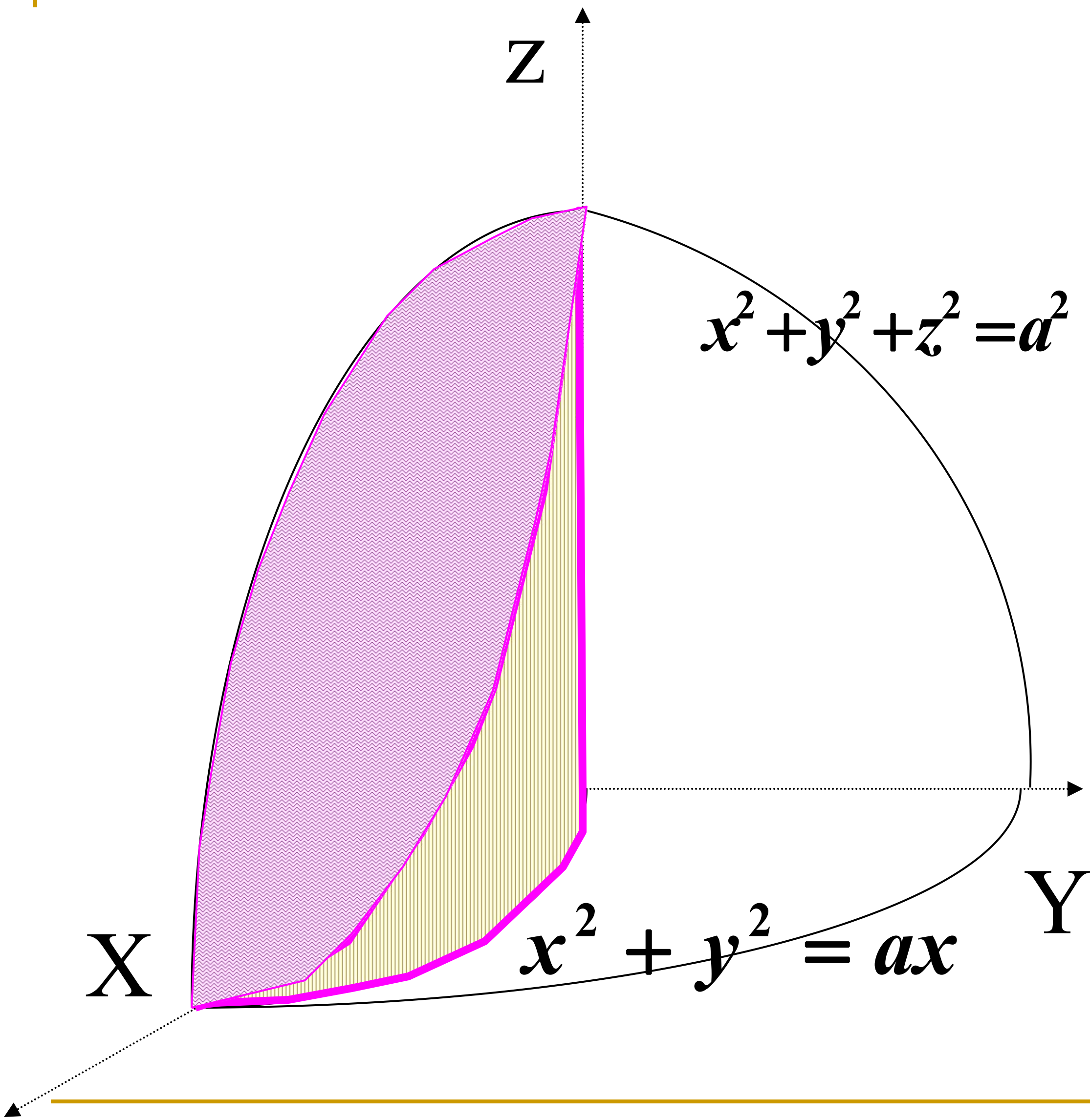
例5 求球面 $x^2+y^2+z^2=a^2$ 与圆柱面 $x^2+y^2=ax$ 所包围的立体体积 V.

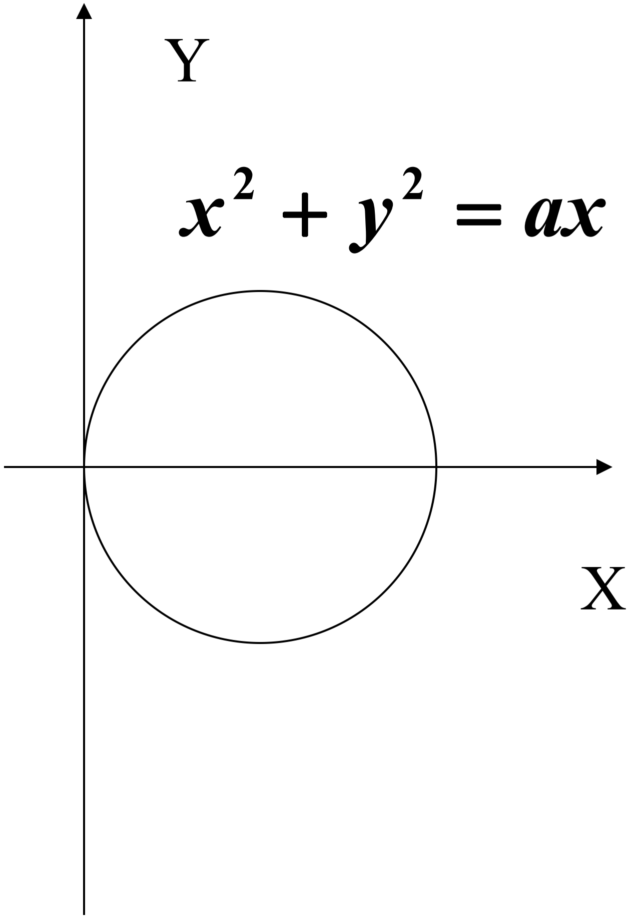
解 $$V=4\iint_D\sqrt{a^2-x^2-y^2}\,dxdy$$

$$=4\int_0^{\frac{\pi}{2}}d\theta\int_0^{a\cos\theta}\sqrt{a^2-\rho^2}\,\rho\,d\rho$$

$$=4\int_0^{\frac{\pi}{2}}[-\frac{1}{2}\cdot\frac{2}{3}(a^2-\rho^2)^{\frac{2}{3}}]_0^{a\cos\theta}d\theta$$

$$=-\frac{4}{3}4\int_0^{\frac{\pi}{2}}[a^3\sin^3\theta-a^3]d\theta=\frac{4a^3}{3}(\frac{\pi}{2}-\frac{2}{3})$$

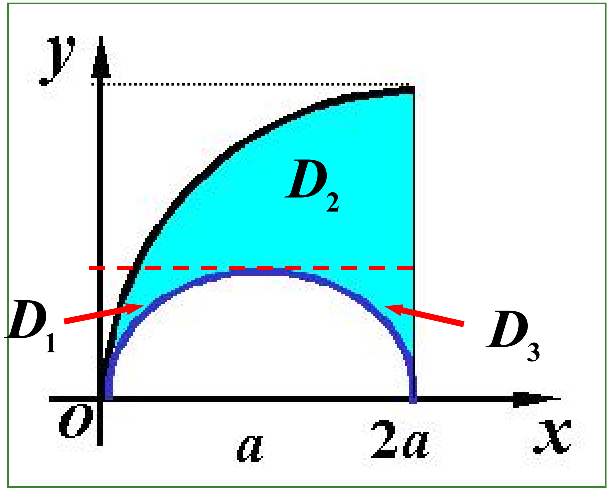

Z
$x^2+y^2+z^2=a^2$
Y
X
$x^2+y^2=ax$


Y
$x^2+y^2=ax$
X

例6 更换积分次序 $I=\int_0^{2a}dx\int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}}f(x,y)dy.\ (a>0)$

解 $D:\begin{cases}0\le x\le 2a,\\ \sqrt{2ax-x^2}\le y\le\sqrt{2ax},\end{cases}$

将积分区域 D 分成 D_1,D_2 及 D_3 三部分，



$D_1:\dfrac{y^2}{2a}\le x\le a-\sqrt{a^2-y^2},$

$0\le y\le a;$

$$D_2: \frac{y^2}{2a} \le x \le 2a,\ a \le y \le 2a;$$

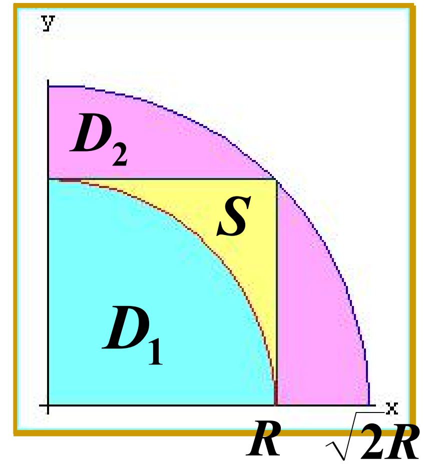
$$D_3: a + \sqrt{a^2 - y^2} \le x \le 2a,$$

$$0 \le y \le a;$$

故 $$I = \int_0^a dy \int_{\frac{y^2}{2a}}^{a-\sqrt{a^2-y^2}} f(x,y)dx$$

$$+\int_a^{2a} dy \int_{\frac{y^2}{2a}}^{2a} f(x,y)dx + \int_0^a dy \int_{a+\sqrt{a^2-y^2}}^{2a} f(x,y)dx$$

例 7 求广义积分 $\int_0^{+\infty} e^{-x^2} dx$.



解 $D_1 = \{(x, y) \mid x^2 + y^2 \le R^2\}$

$D_2 = \{(x, y) \mid x^2 + y^2 \le 2R^2\}$

$S = \{(x, y) \mid 0 \le x \le R, 0 \le y \le R\}$

$\{x \ge 0, y \ge 0\}$ 显然有 $D_1 \subset S \subset D_2$

$\because \quad e^{-x^2-y^2} > 0,$

$$\therefore \quad \iint_{D_1} e^{-x^2-y^2} dxdy \le \iint_{S} e^{-x^2-y^2} dxdy \le \iint_{D_2} e^{-x^2-y^2} dxdy.$$

又$\because\ I=\iint\limits_{S} e^{-x^2-y^2}dxdy$

$$=\int_0^R e^{-x^2}dx\int_0^R e^{-y^2}dy\ =(\int_0^R e^{-x^2}dx)^2;$$

$$I_1=\iint\limits_{D_1} e^{-x^2-y^2}dxdy$$

$$=\int_0^{\frac{\pi}{2}}d\theta\int_0^R e^{-\rho^2}\rho d\rho\ =\frac{\pi}{4}(1-e^{-R^2});$$

同理$I_2=\iint\limits_{D_2} e^{-x^2-y^2}dxdy=\frac{\pi}{4}(1-e^{-2R^2});$

$\because I_1 \le I \le I_2,$

$$\therefore \frac{\pi}{4}(1-e^{-R^2}) \le \left(\int_0^R e^{-x^2}dx\right)^2 \le \frac{\pi}{4}(1-e^{-2R^2});$$

当 $R \to +\infty$ 时 $I_1 \to \frac{\pi}{4}$, $I_2 \to \frac{\pi}{4}$,

当 $R \to +\infty$ 时 $I \to \frac{\pi}{4}$, 即 $\left(\int_0^{+\infty} e^{-x^2}dx\right)^2 = \frac{\pi}{4}$,

所求广义积分 $\int_0^{+\infty} e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$.

课堂练习

一、交换积分次序：且化为极坐标形式的二次积分

$$I=\int_3^4 dx\int_0^{\sqrt{4x-x^2}} f(x,y)dy+\int_{\frac{9}{4}}^3 dx\int_{\sqrt{9-x^2}}^{\sqrt{4x-x^2}} f(x,y)dy$$

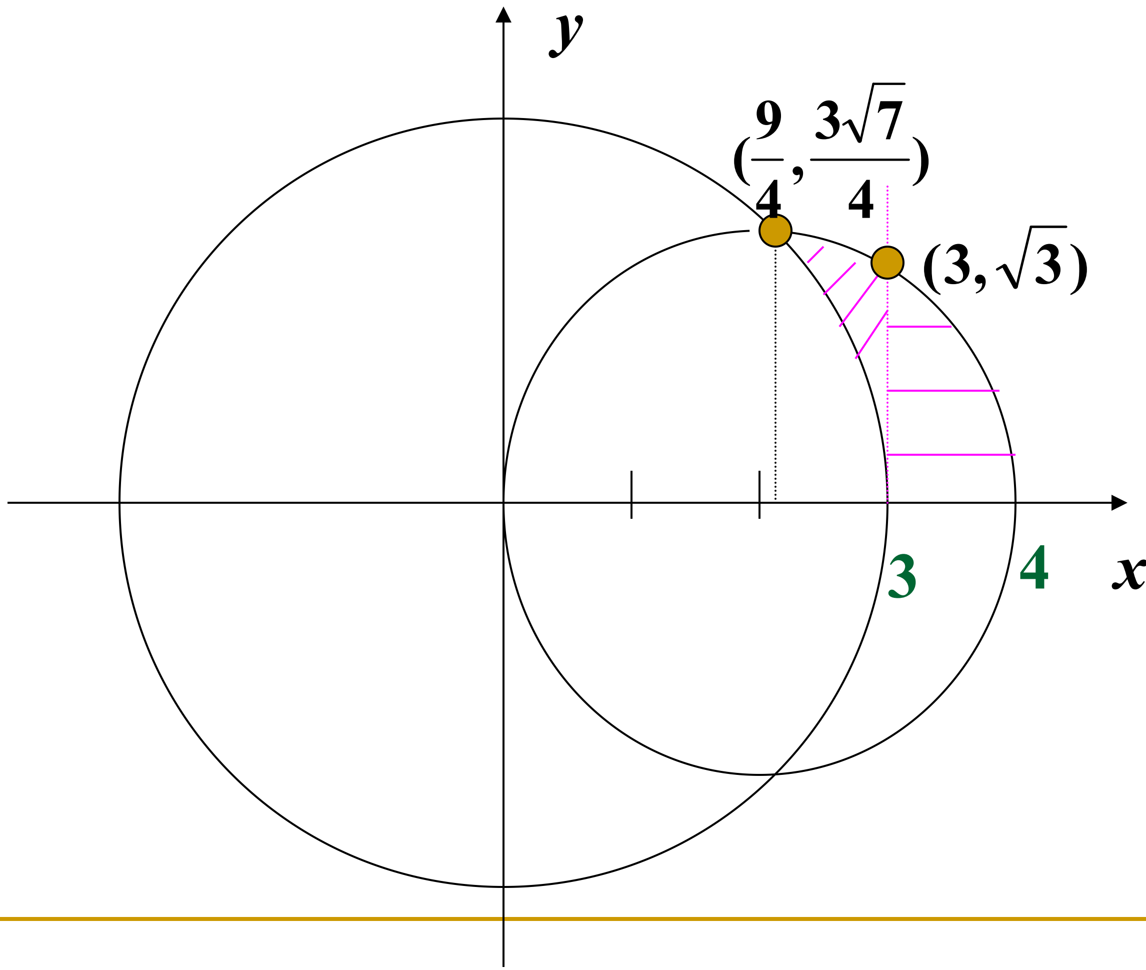
解

$$D_1:\begin{cases}\frac{9}{4}\le x\le 3\\ \sqrt{9-x^2}\le y\le\sqrt{4x-x^2}\end{cases},\quad D_2:\begin{cases}3\le x\le 4\\ 0\le y\le\sqrt{4x-x^2}\end{cases}$$

$$D_1+D_2\text{可写成：}\begin{cases}0\le y\le\frac{3\sqrt{7}}{4}\\ 0\le y\le 2+\sqrt{4-y^2}\end{cases},\text{交点：}\left(\frac{9}{4},\frac{3\sqrt{7}}{4}\right)$$

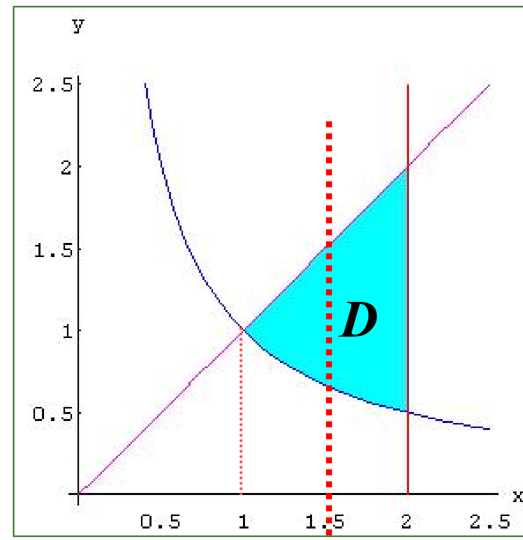
$$I=\int_0^{\frac{3\sqrt{7}}{4}} dy\int_{\sqrt{9-y^2}}^{2+\sqrt{4-y^2}} f(x,y)dx$$

$$I = \int_0^{arctg\frac{\sqrt{7}}{3}} d\theta \int_3^{4\cos\theta} f(\rho\cos\theta, \rho\sin\theta)\rho d\rho$$



二、计算下列各题

1.计算 $\iint\limits_D \frac{x^2}{y^2}d\sigma$. 其中 D 由 $y=x, y=\frac{1}{x}, x=2$ 围成.



解 X-型 $D: \frac{1}{x} \le y \le x, 1 \le x \le 2.$

$$\iint\limits_D \frac{x^2}{y^2}d\sigma = \int_1^2 dx \int_{\frac{1}{x}}^{x} \frac{x^2}{y^2}dy$$

$$= \int_1^2 \left(-\frac{x^2}{y}\right)\Bigg|_{\frac{1}{x}}^{x} dx = \int_1^2 (x^3 - x)dx = \frac{9}{4}$$ 。

$$2.\iint_D |xy|dxdy \quad D: x^2+y^2 \le a^2$$

解法1 $$\iint_D |xy|dxdy = 4\iint_{D_1} xydxdy$$

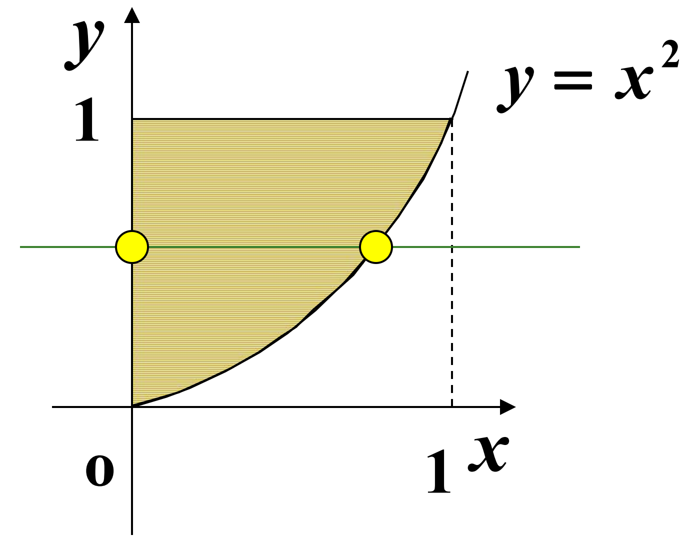
$$= 4\iint_{D_1} xydxdy = 4\int_0^{\frac{\pi}{2}} d\theta \int_0^a \rho^3 \sin\theta\cos\theta d\rho$$

$$= a^4\int_0^{\frac{\pi}{2}} \frac{1}{2}\sin 2\theta d\theta = \frac{a^4}{4}\cdot[-\cos 2\theta]_0^{\frac{\pi}{2}} \quad = \frac{a^4}{2}$$

解法2 $$\iint_D |xy|dxdy = 4\iint_{D_1} xydxdy$$

$$= 4\int_0^a dx \int_0^{\sqrt{a^2-x^2}} xydy = \frac{a^4}{2}$$

3.计算$\int_0^1 dx\int_{x^2}^1 \frac{xy}{\sqrt{1+y^3}}dy$



解 原式$=\int_0^1 dy\int_0^{\sqrt{y}} \frac{xy}{\sqrt{1+y^3}}dx$

$$D:\begin{matrix}0\le x\le 1\\ x^2\le y\le 1\end{matrix}$$

$$=\int_0^1 \frac{y}{\sqrt{1+y^3}}[\frac{x^2}{2}]_0^{\sqrt{y}}dy$$

$$=\int_0^1 \frac{y^2}{2\sqrt{1+y^3}}dy=\int_0^1 \frac{1}{6}\frac{1}{\sqrt{1+y^3}}dy^3$$

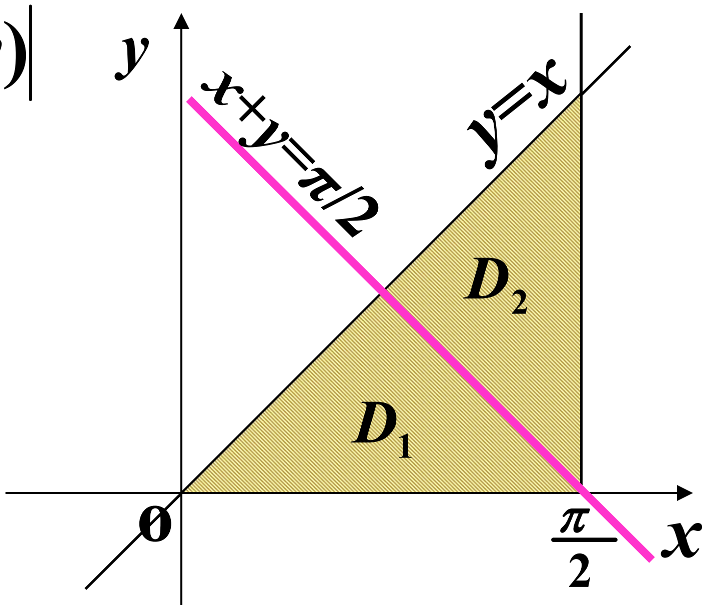
$$=\frac{1}{3}[\sqrt{1+y^3}]_0^1$$

4.计算：$I=\iint_D \sqrt{1-\sin^2(x+y)}d\sigma$

D:由$y=x, y=0, x=\dfrac{\pi}{2}$所围区域。

解 $\because \sqrt{1-\sin^2(x+y)}=\left|\cos(x+y)\right|$

$$\begin{cases} \geq 0 & 0 \leq x+y \leq \dfrac{\pi}{2} \\ < 0 & \dfrac{\pi}{2} < x+y \leq \pi \end{cases}$$



$\therefore D = D_1 + D_2$

$$I=\iint\limits_{D}\left|\cos(x+y)\right|d\sigma$$

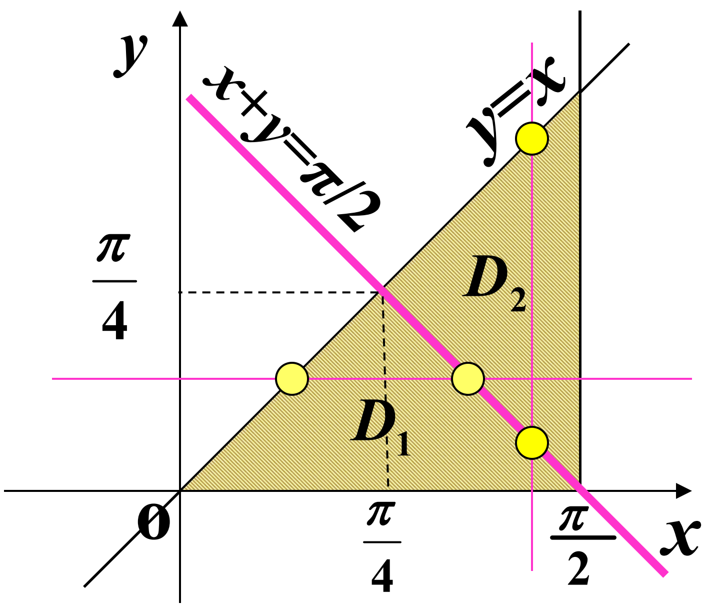
$$=\iint\limits_{D_1}\cos(x+y)d\sigma-\iint\limits_{D_2}\cos(x+y)d\sigma$$

$$=\int_0^{\frac{\pi}{4}}dy\int_y^{\frac{\pi}{2}-y}\cos(x+y)dx$$

$$-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}dx\int_{\frac{\pi}{2}-x}^{x}\cos(x+y)dy$$

$$=\int_0^{\frac{\pi}{4}}(1-\sin 2y)dy$$

$$-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(\sin 2x-1)dx=\frac{\pi}{2}-1$$



三、证明:

$$\int_a^b dx\int_a^x (x-y)^{n-2} f(y)dy = \frac{1}{n-1}\int_a^b (b-y)^{n-1} f(y)dy.$$

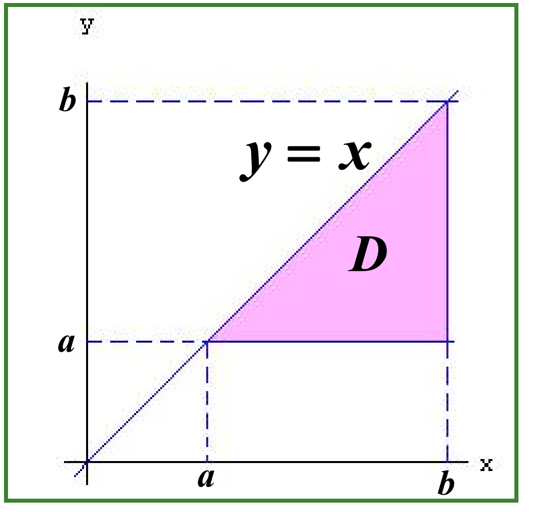
证

$$\int_a^b dx\int_a^x (x-y)^{n-2} f(y)dy$$

$$= \int_a^b dy\int_y^b (x-y)^{n-2} f(y)dx$$

$$= \int_a^b f(y)dy[\frac{1}{n-1}(x-y)^{n-1}]_y^b$$

$$= \frac{1}{n-1}\int_a^b (b-y)^{n-1} f(y)dy.$$



四、设有一平面$z=ax+by+c$在这平面上取一块区域S,设S在xoy面上的投影域为D,试证明:D为底,S为顶的柱体体积$V=H\sigma$.(其中H为D的质量中心处的高)

解

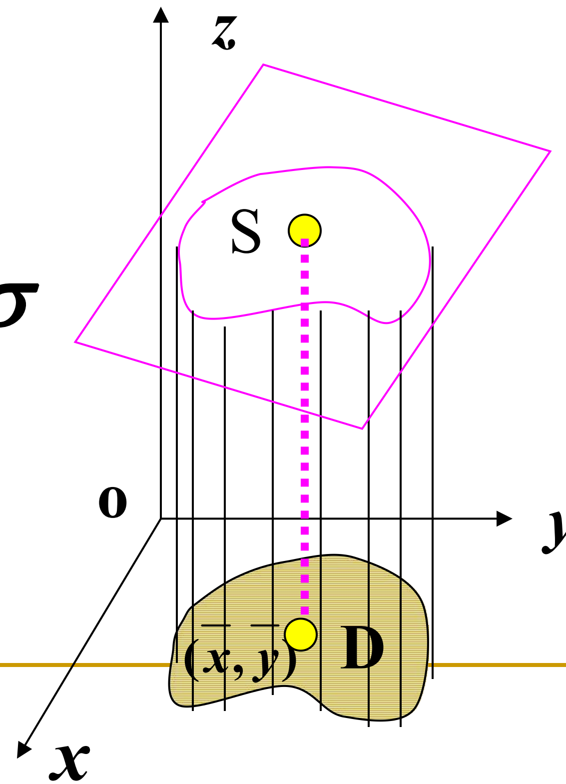
$$H=a\bar{x}+b\bar{y}+c$$

$$V=\iint\limits_D (ax+by+c)d\sigma$$

$$=\iint\limits_D axd\sigma+\iint\limits_D byd\sigma+\iint\limits_D cd\sigma$$

$$=a\bar{x}\sigma+b\bar{y}\sigma+c\sigma$$

$$=(a\bar{x}+b\bar{y}+c)\sigma=H\sigma$$



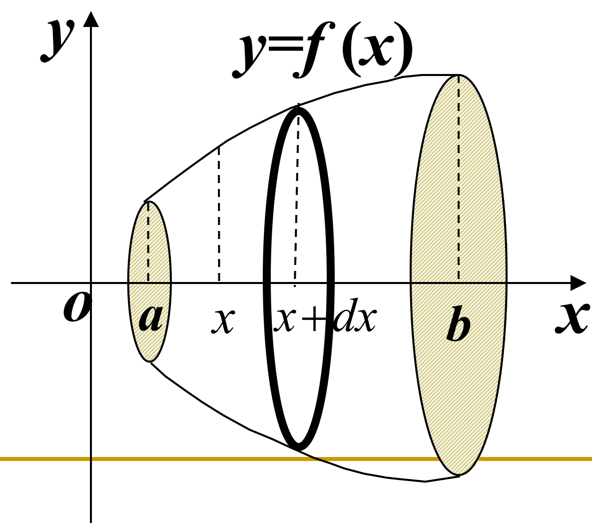
五：求证：曲线$y = f(x) \quad a \le x \le b \quad (f(x) \ge 0)$

绕x轴旋转一周而成的曲面面积为

$$A = \int_a^b 2\pi f(x)\sqrt{1 + f'(x)^2}\,dx$$

并求正弦曲线$y = \sin x$由$x_1 = 0$，到$x_2 = \dfrac{\pi}{2}$

一段绕x轴旋转一周而成的曲面面积



证明　旋转面方程为

$$y^2 + z^2 = f^2(x)$$

$$z = \pm\sqrt{f^2(x) - y^2}$$

$$D: \begin{array}{c} a \le x \le b \\ -f(x) \le y \le f(x) \end{array}$$

$$\frac{\partial z}{\partial x}=\frac{ff'}{\sqrt{f^2-y^2}} \qquad \frac{\partial z}{\partial y}=\frac{-y}{\sqrt{f^2-y^2}}$$

$$\sqrt{1+z_x^2+z_y^2}=\frac{f\sqrt{1+f'^2}}{\sqrt{f^2-y^2}}$$

$$A=2\iint_D\sqrt{1+z_x^2+z_y^2}\,dxdy=2\iint_D\frac{f\sqrt{1+f'^2}}{\sqrt{f^2-y^2}}\,dxdy$$

$$=2\int_a^b dx\int_{-f}^{f}\frac{f(x)\sqrt{1+f'^2(x)}}{\sqrt{f^2-y^2}}\,dy$$

$$=2\int_a^b[\arcsin\frac{y}{f}]_{-f}^{f}\,f\sqrt{1+f'^2}\,dx=2\pi\int_a^b f(x)\sqrt{1+f'^2(x)}\,dx$$

解 $A=\int_0^{\frac{\pi}{2}} 2\pi\sin x\sqrt{1+\cos^2 x}dx$

$$=-2\pi\left[\frac{\cos x\sqrt{1+\cos^2 x}}{2}+\frac{1}{2}\ln(\cos x+\sqrt{1+\cos^2 x})\right]_0^{\frac{\pi}{2}}$$

$$=\pi(\sqrt{2}+\ln(1+\sqrt{2}))]$$

又解 旋转面方程为$\sqrt{y^2+z^2}=\sin x$

$x=\arcsin\sqrt{y^2+z^2} \quad D: y^2+z^2\le 1$

$$\frac{\partial x}{\partial y}=\frac{1}{\sqrt{1-y^2-z^2}}\frac{y}{\sqrt{y^2+z^2}} \qquad \frac{\partial x}{\partial z}=\frac{1}{\sqrt{1-y^2-z^2}}\cdot\frac{z}{\sqrt{y^2+z^2}}$$

$$\left(\frac{\partial x}{\partial y}\right)^2+\left(\frac{\partial x}{\partial z}\right)^2+1=\frac{1}{1-y^2-z^2}+1=\frac{2-y^2-z^2}{1-y^2-z^2}$$

三重积分计算与应用

习题课

主要内容

举　　例

课堂练习

主要内容

定义 $\iiint\limits_{\Omega} f(x,y,z)dv=\lim\limits_{\lambda\to 0}\sum\limits_{i=1}^{n} f(x_i,y_i,z_i)\Delta v_i$

(与分法无关、点的取法　无关）– 注意物理意义

性质　与二重积分相类似的性质（线性性、对区域的可加性、比较性、估值、中值）

★计算　选择坐标 – 化为三次定积分(累次积分）

(1)利用直角坐标 $\begin{cases} 先"\int"后"\iint" \\ 先"\iint"后"\int" \end{cases}$

— 注意 Ω 须满足一定的条件

先"$\iint$"后"$\int$ "计算三重积分的方法

如被积函数只依赖于一个变元，如$f(x,y,z)=f(z)$,并当 z=常数的平面与Ω相截时，其截面积 S_z易求,则可将三重积分的计算转化为 先"$\iint$"后"$\int$ "

设在Ω中，$z_{min}=c, z_{max}=d$中，用平面$Z=z$截Ω得其平面区域S_z (截面积也记S_z）, 在xoy面上的投影区域记为D_z,则

$$\iiint\limits_{\Omega} f(z)dxdydz = \int_c^d f(z)dz \iint\limits_{D_z} dxdy = \int_c^d f(z)S_z dz$$

(2)利用柱坐标 $: dv = r dr d\theta dz$

$$x = r\cos\theta,\ y = r\sin\theta,\ z = z$$

(3)利用球坐标 $: dv = r^2 \sin\varphi dr d\varphi d\theta$

$$x = r\sin\varphi\cos\theta, y = r\sin\varphi\sin\theta, z = r\cos\varphi$$

(4)利用变量代换： $(x,y,z)\in\Omega \Rightarrow (u,v,w)\in\Omega'$

令$x = x(u,v,w)$,

$$y = y(u,v,w), \quad J = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix} \quad dv = |J| du dv dw$$

$z = z(u,v,w)$

$$\iiint_{\Omega} f(x,y,z)dv =$$

$$= \iiint_{\Omega'} f(x(u,v,w), y(u,v,w), z(u,v,w))|J| du dv dw$$

(5)三重积分的对称性

若$\Omega=\Omega_1+\Omega_2$且Ω_1与Ω_2关于xoy面对称，

$f(x,y,z)$为连续函数，

则$\iiint\limits_{\Omega} f(x,y,z)dv$

$$=\begin{cases} 2\iiint\limits_{\Omega_1} f(x,y,z)dv & \text{当} f(x,y,z)=f(x,y,-z) \\ 0 & \text{当} f(x,y,-z)=-f(x,y,z) \end{cases}$$

应用

几何应用　求空间区域Ω的体积：　$v=\iiint\limits_{\Omega} dv$

物理应用

1.求非均匀分布空间物体Ω的质量M：

$$M=\iiint\limits_{\Omega}\rho(x,y,z)dv$$

2.求非均匀分布空间物体Ω的重心

$$\bar{x}=\frac{\iiint\limits_{\Omega} x\rho(x,y,z)dv}{M},\ \bar{y}=\frac{\iiint\limits_{\Omega} y\rho(x,y,z)dv}{M},$$

$$\bar{z}=\frac{\iiint\limits_{\Omega} z\rho(x,y,z)dv}{M}$$

3.求非均匀分布空间物体Ω绕坐标轴的转动惯量

$$I_x = \iiint_{\Omega} (y^2 + z^2)\rho(x, y, z)dv$$

$$I_y = \iiint_{\Omega} (x^2 + z^2)\rho(x, y, z)dv$$

$$I_z = \iint_{D} (x^2 + y^2)\rho(x, y, z)dv$$

4.求非均匀分布空间物体Ω对物体外一质点的引力

举　　例

例1　求 $\iiint\limits_{\Omega} e^{|z|}dv$，其中 Ω： $x^2+y^2+z^2\leq 1$

解法一　$\iiint\limits_{\Omega} e^{|z|}dv=\int_{-1}^{1}e^{|z|}dz\iint\limits_{D_z}dxdy$　　$D_z: x^2+y^2\leq 1-z^2$

$$=\pi\int_{-1}^{1}e^{|z|}(1-z^2)dz=2\pi\int_0^1 e^z(1-z^2)dz=2\pi$$

解法二 $\Omega_1: x^2+y^2+z^2 \le 1$ 及 $z \ge 0$

$D: x^2+y^2 \le 1$

$$\iiint\limits_{\Omega} e^{|z|} dv = 2\iiint\limits_{\Omega_1} e^{z} dv$$

$$= 2\iint\limits_{D} dxdy \int_0^{\sqrt{1-x^2-y^2}} e^z dz = 2\iint\limits_{D} (e^{\sqrt{1-x^2-y^2}} - 1) dxdy$$

$$= 2\int_0^{2\pi} d\theta \int_0^1 (e^{\sqrt{1-\rho^2}} - 1)\rho d\rho$$

$$= 4\pi \int_0^1 e^{\sqrt{1-\rho^2}} \rho d\rho - 2\pi$$

$$\overset{\text{令}\sqrt{1-\rho^2}=t}{=} 4\pi \int_0^1 e^t t dt - 2\pi = 2\pi$$

解法三 $\iiint\limits_{\Omega} e^{|z|}dv = 2\iiint\limits_{\Omega_1} e^{z}dv$

$$\Omega_1 : x^2 + y^2 + z^2 \le 1 \text{及} z \ge 0$$

$$\text{原式} = 2\iiint\limits_{\Omega_1} e^{r\cos\varphi} r^2 \sin\varphi dr d\varphi d\theta$$

$$= 2\int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 e^{r\cos\varphi} r^2 \sin\varphi dr$$

$$= 4\pi \int_0^{\frac{\pi}{2}} \sin\varphi d\varphi \int_0^1 e^{r\cos\varphi} r^2 dr$$

$$= 4\pi \int_0^{\frac{\pi}{2}} \sin\varphi e^{\cos\varphi} [\frac{1}{\cos\varphi} - \frac{2}{\cos^2\varphi} - \frac{2}{\cos^3\varphi}] d\varphi$$

$$= \cdots = 2\pi$$

例2 求$\iiint\limits_{\Omega}\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)dv$ 其中Ω: $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\leq 1$

解法一 令$x = ar\sin\varphi\cos\theta, y = br\sin\varphi\sin\theta, z = cr\cos\varphi$

$$\Omega \to \Omega': r \leq 1,$$

$$J = \begin{vmatrix} x_r & x_\varphi & x_\theta \\ y_r & y_\varphi & y_\theta \\ z_r & z_\varphi & z_\theta \end{vmatrix}$$

$$= \begin{vmatrix} a\sin\varphi\cos\theta & ar\cos\varphi\cos\theta & -ar\sin\varphi\sin\theta \\ b\sin\varphi\sin\theta & br\cos\varphi\sin\theta & br\sin\varphi\cos\theta \\ c\cdot\cos\varphi & -cr\sin\varphi & 0 \end{vmatrix}$$

$$= abcr^2\sin\varphi$$

$$\iiint_{\Omega}\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)dv$$

$$=\iiint_{r\leq 1} r^2 \cdot abcr^2 \sin\varphi\, dr\, d\varphi\, d\theta$$

$$=\int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^1 abcr^4 \sin\varphi\, dr = \frac{4}{5}abc\pi$$

解法二

$$D_x: \frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1-\frac{x^2}{a^2},$$

$$S_x = \pi bc\left(1-\frac{x^2}{a^2}\right)$$

$$\iiint_{\Omega}\left(\frac{x^2}{a^2}\right)dv = \int_{-a}^{a}\frac{x^2}{a^2}dx\iint_{D_x}dydz$$

$$= \int_{-a}^{a}\frac{x^2}{a^2}\pi bc\left(1-\frac{x^2}{a^2}\right)dx$$

$$= 2\int_{0}^{a}\frac{x^2}{a^2}\pi bc\left(1-\frac{x^2}{a^2}\right)dx$$

$$= 2\pi bc\left[\frac{x^3}{3a^2}-\frac{x^5}{5a^4}\right]_0^a = \frac{4}{15}\pi abc$$

同理，$\iiint_{\Omega}\left(\frac{y^2}{b^2}\right)dv = \iiint_{\Omega}\left(\frac{z^2}{c^2}\right)dv = \frac{4}{15}\pi abc$

$$\therefore \iiint_{\Omega}\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)dv$$

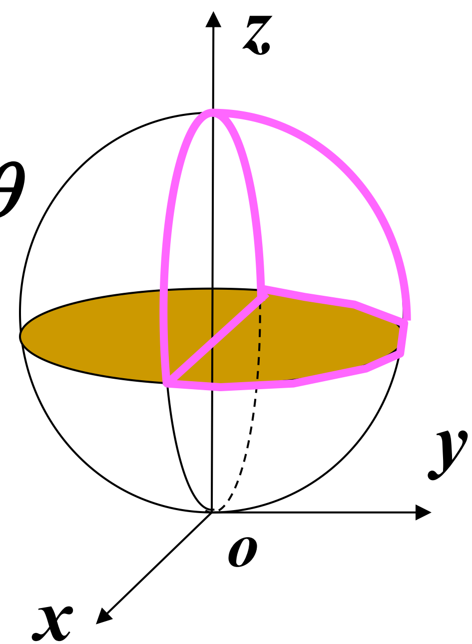
$$= \frac{4}{5}\pi abc$$

例3 计算：$I=\int_{-1}^{1}dx\int_{0}^{\sqrt{1-x^2}}dy\int_{1}^{1+\sqrt{1-x^2-y^2}}\frac{dz}{\sqrt{x^2+y^2+z^2}}$

解

$$\Omega:\begin{cases}-1\le x\le 1\\ 0\le y\le\sqrt{1-x^2}\\ 1\le z\le 1+\sqrt{1-x^2-y^2}\end{cases}$$ —为四分之一球

用球坐标做： $dv=r^2\sin\varphi drd\varphi d\theta$



$$\Omega':\begin{cases}0\le\theta\le\pi\\ 0\le\varphi\le\dfrac{\pi}{4}\\ 1/\cos\varphi\le r\le 2\cos\varphi\end{cases}$$

$$\therefore I=\int_0^{\pi}d\theta\int_0^{\frac{\pi}{4}}d\varphi\int_{\frac{1}{\cos\varphi}}^{2\cos\varphi}\frac{r^2\sin\varphi}{r}dr=\frac{\pi}{6}(7-4\sqrt{2})$$

例4 设$f(x)$为连续函数，$F(t)=\iiint\limits_{\Omega}[z^2+f(x^2+y^2)]dv$

其中Ω为 $0\le z\le h, x^2+y^2\le t^2 (t>0)$,求$\dfrac{dF}{dt}$

解 $\Omega:\begin{cases} 0\le\theta\le 2\pi \\ 0\le\rho\le t \\ 0\le z\le h \end{cases}$ $\quad dv=\rho d\rho d\theta dz$

$$F(t)=\int_0^{2\pi}d\theta\int_0^t d\rho\int_0^h[z^2+f(\rho^2)]\rho dz$$

$$=2\pi\int_0^t[hf(\rho^2)\rho+\frac{h^3}{3}\rho]d\rho$$

$$\therefore F'(t)=2\pi ht[f(t^2)+\frac{h^2}{3}]$$

例5 将三次积分 $I=\int_0^1 dx\int_x^1 dy\int_x^y f(x,y,z)dz$

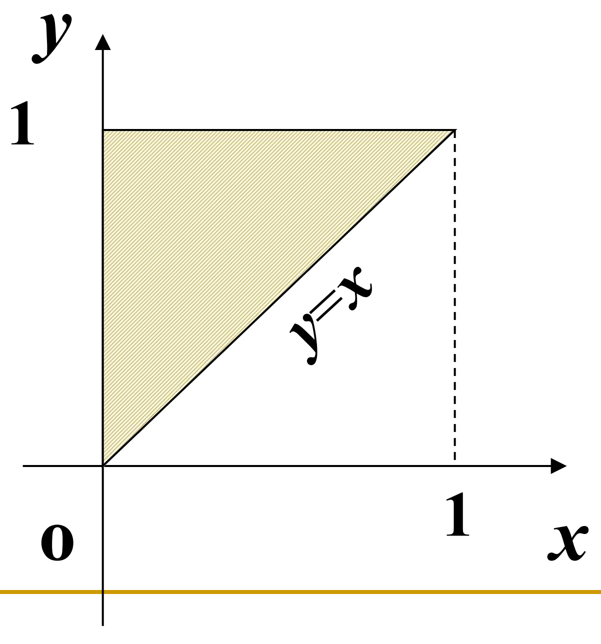
改变积分次序为先对 x,再对 y,最后对 z

则 $I=$ $\underline{\int_0^1 dz\int_z^1 dy\int_0^z f(x,y,z)dx}$

解 $I=\iint\limits_{D_{xy}} dxdy\int_x^y f(x,y,z)dz$

$$=\int_0^1 dy\int_0^y dx\int_x^y f(x,y,z)dz$$

$$D_{xy}:\begin{cases}0\le x\le 1\\ x\le y\le 1\end{cases}$$

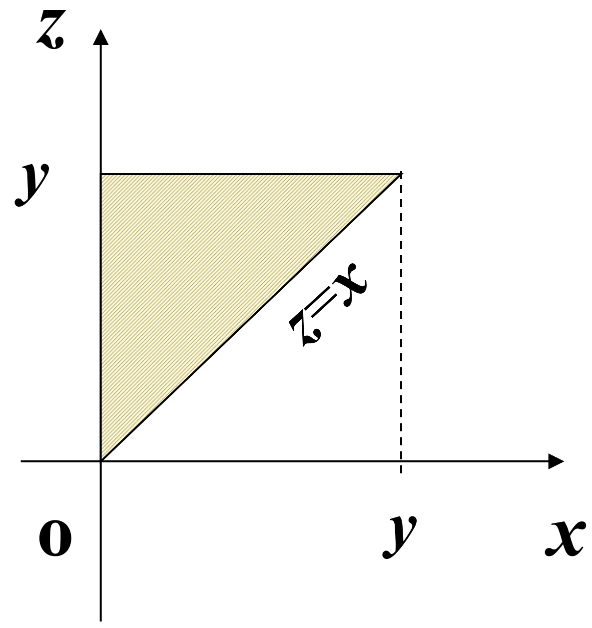


$$I=\iint_{D_{xy}} dxdy\int_x^y f(x,y,z)dz$$

$$=\int_0^1 dy\int_0^y dx\int_x^y f(x,y,z)dz$$

$$=\int_0^1 dy \iint_{D(y)} f(x,y,z)dxdz$$

$$=\int_0^1 dy\int_0^y dz\int_0^z f(x,y,z)dx$$



$$D(y):\begin{cases}0\le x\le y\\ x\le z\le y\end{cases}(y\text{视为常数})$$

$$I=\iint_{D_{xy}} dxdy\int_x^y f(x,y,z)dz$$

$$=\int_0^1 dy\int_0^y dx\int_x^y f(x,y,z)dz$$

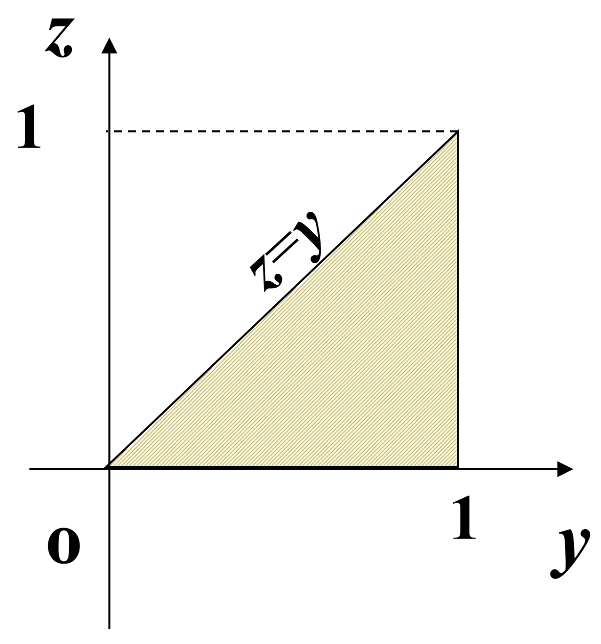
$$=\int_0^1 dy\iint_{D(y)} f(x,y,z)dxdz$$

$$=\int_0^1 dy\int_0^y dz\int_0^z f(x,y,z)dx$$

$$=\iint_{D_{yz}} dydz\int_0^z f(x,y,z)dx$$

$$=\int_0^1 dz\int_z^1 dy\int_0^z f(x,y,z)dx$$

$$D_{yz}:\begin{cases}0\le y\le 1\\ 0\le z\le y\end{cases}$$



课堂练习

一、计算题

1. $\iiint\limits_{\Omega}(x+y+z)dv$，其中$\Omega$为$x^2+y^2-z^2=0$与$z=1$所围区域。　　(答案：$\frac{\pi}{4}$)

2. $\iiint\limits_{\Omega}(x+y+z)^2 dv$

$\Omega: z \geq x^2+y^2, x^2+y^2+z^2 \leq 2$

（答案：$(\frac{8}{5}\sqrt{2}-\frac{89}{60})\pi$）

3. $\iiint\limits_{\Omega} xy\,dx\,dy\,dz$

$\Omega: x^2+y^2+z^2 \le 4,$

$x^2+y^2+(z-2)^2 \le 4,$

$x \ge 0, y \ge 0$

（答案：$\frac{53}{60}$）

4. $\iiint\limits_{\Omega} \left|1-\sqrt{x^2+y^2+z^2}\right| dv,$

Ω：由$z=\sqrt{x^2+y^2}, z=1$所围。

(答案：$\frac{\pi}{6}(\sqrt{2}-1)$)

1. $\iiint\limits_{\Omega}(x+y+z)dv$,其中$\Omega$为$x^2+y^2-z^2=0$与$z=1$所围区域。

解法一　由对称性　$I=\iiint\limits_{\Omega} zdxdydz$

由截面法　$D_z: x^2+y^2 \leq z^2$

$$I=\int_0^1 zdz\iint\limits_{D_z} dxdy=\int_0^1 z\pi z^2 dz=\frac{\pi}{4}$$

解法二　由对称性　$I = \iiint\limits_{\Omega} z\,dxdydz$

$$I = \iint\limits_{D} dxdy \int_{\sqrt{x^2+y^2}}^{1} z\,dz \qquad D: x^2 + y^2 \le 1$$

$$= \iint\limits_{D} \frac{1}{2}(1 - x^2 - y^2)dxdy$$

$$= \frac{1}{2}\int_0^{2\pi} d\theta \int_0^1 (1-\rho^2)\rho d\rho$$

$$= \pi[\frac{\rho^2}{2} - \frac{\rho^4}{4}]_0^1 = \frac{\pi}{4}$$

解法三　由对称性　$I=\iiint_{\Omega} z\,dxdydz$

用球坐标做

$$I=\iiint_{\Omega} z\,dxdydz=\iiint_{\Omega} r\cos\varphi r^{2}\sin\varphi\, dr d\varphi d\theta$$

$$=\int_0^{2\pi}d\theta\int_0^{\frac{\pi}{4}}d\varphi\int_0^{\frac{1}{\cos\varphi}}r^3\cos\varphi\sin\varphi dr$$

$$=\frac{\pi}{2}\int_0^{\frac{\pi}{4}}-\frac{1}{\cos^3\varphi}d\cos\varphi=\frac{\pi}{4}[\frac{1}{\cos^2\varphi}]_0^{\frac{\pi}{4}}=\frac{\pi}{4}$$

2. $\iiint\limits_{\Omega}(x+y+z)^2dv$

$$\Omega: z \ge x^2+y^2, x^2+y^2+z^2 \le 2$$

解法一 $(x+y+z)^2 = x^2+y^2+z^2+2xy+2yz+2zx$

从 $\begin{cases} x^2+y^2=z \\ x^2+y^2+z^2=2 \end{cases}$ 消去 z 求得投影区域 D: $x^2+y^2 \le 1$

$$\iiint\limits_{\Omega}(x+y+z)^2dv = \iiint\limits_{\Omega}(x^2+y^2+z^2)dv$$

$$= \iiint\limits_{\Omega}(\rho^2+z^2)\rho d\rho d\theta dz = \int_0^{2\pi}d\theta\int_0^1 \rho d\rho\int_{r^2}^{\sqrt{2-r^2}}(\rho^2+z^2)dz$$

$$= (\frac{8}{5}\sqrt{2}-\frac{89}{60})\pi$$

解法二 $\displaystyle\iiint\limits_{\Omega}(x+y+z)^2dv=\iiint\limits_{\Omega}(x^2+y^2+z^2)dv$

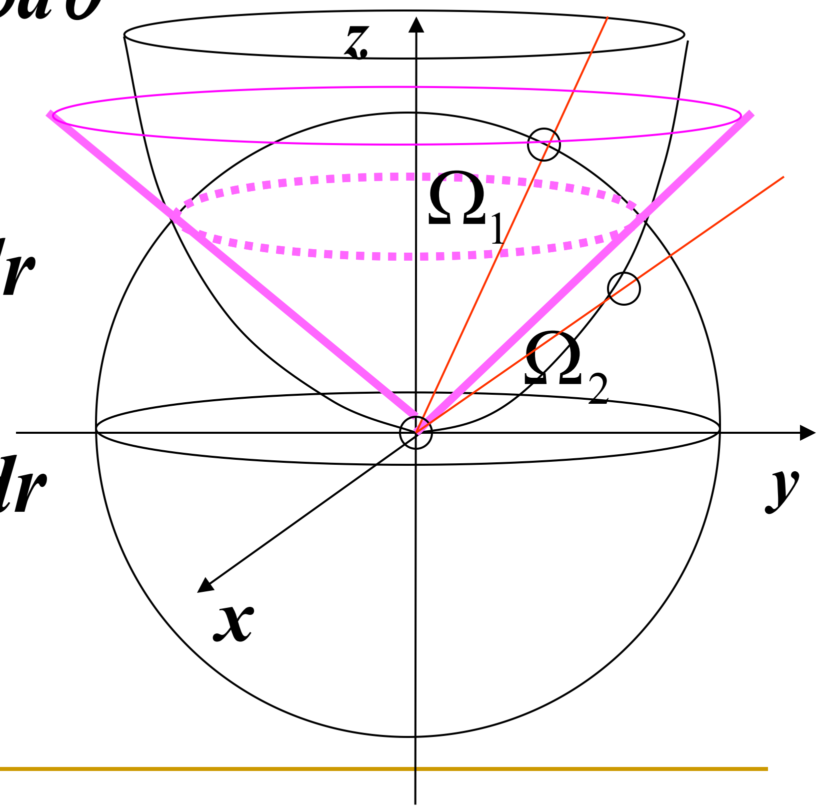
$$=\iiint\limits_{\Omega} r^2r^2\sin\varphi drd\varphi d\theta$$

$$=\iiint\limits_{\Omega_2}\quad+\iiint\limits_{\Omega_2} r^2r^2\sin\varphi drd\varphi d\theta$$

$$=\int_0^{2\pi}d\theta\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\sin\varphi d\varphi\int_0^{\frac{\cos\varphi}{\sin^2\varphi}}r^4dr$$

$$+\int_0^{2\pi}d\theta\int_0^{\frac{\pi}{4}}d\varphi\int_0^{\sqrt{2}}r^4\sin\varphi\, dr$$

$$=(\frac{8}{5}\sqrt{2}-\frac{89}{60})\pi$$



3. $\iiint\limits_{\Omega} xydxdydz$

$\Omega: x^2+y^2+z^2 \le 4,$

$x^2+y^2+(z-2)^2 \le 4,$

$x \ge 0, y \ge 0$

解 从$\begin{cases} x^2+y^2+z^2=4 \\ x^2+y^2+(z-2)^2=4 \end{cases}$

消去z得$D: x^2+y^2 \le 3$

$x^2+y^2 \le 3, x \ge 0, y \ge 0$

$$\iiint\limits_{\Omega} xydxdydz = \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt{3}} \rho d\rho \int_{2-\sqrt{4-\rho^2}}^{\sqrt{4-\rho^2}} \rho^2 \cos\theta \sin\theta dz$$

$$= \int_0^{\frac{\pi}{2}} \cos\theta \sin\theta d\theta \int_0^{\sqrt{3}} \rho\rho^2 (\sqrt{4-\rho^2} - 2 + \sqrt{4-\rho^2}) d\rho = \frac{53}{60}$$

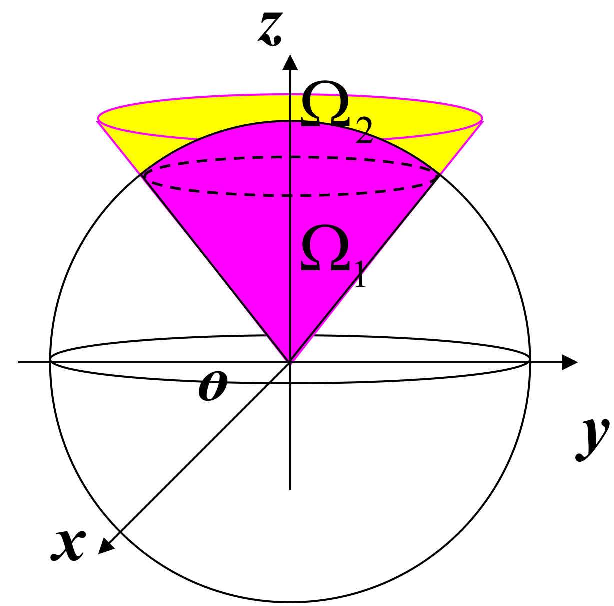
4. $\iiint\limits_{\Omega}\left|1-\sqrt{x^2+y^2+z^2}\right|dv,\Omega:$由$z=\sqrt{x^2+y^2}, z=1$所围。

解 $\left|1-\sqrt{x^2+y^2+z^2}\right|=\begin{cases}1-\sqrt{x^2+y^2+z^2} & x^2+y^2+z^2\le 1\\ \sqrt{x^2+y^2+z^2}-1 & x^2+y^2+z^2\ge 1\end{cases}$

用球：$x^2+y^2+z^2=1$

将Ω分成二部分：

$\Omega=\Omega_1+\Omega_2$



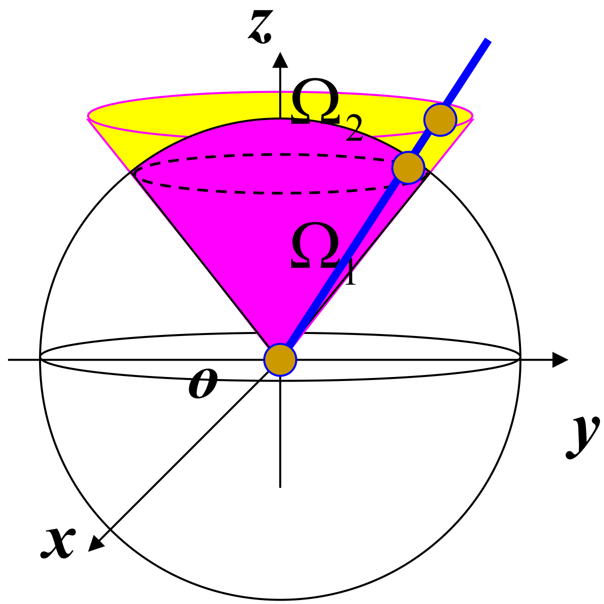
$\iiint\limits_{\Omega}\left|1-\sqrt{x^2+y^2+z^2}\right|dv=$

$=\iiint\limits_{\Omega_1}[1-\sqrt{x^2+y^2+z^2}]dv+\iiint\limits_{\Omega_2}[\sqrt{x^2+y^2+z^2}-1]dv$

$$=\int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^1 (1-r) r^2 \sin\varphi dr$$

$$+\int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_1^{\frac{1}{\cos\varphi}} (r-1) r^2 \sin\varphi d\varphi$$

$$=\frac{\pi}{6}(\sqrt{2}-1)$$



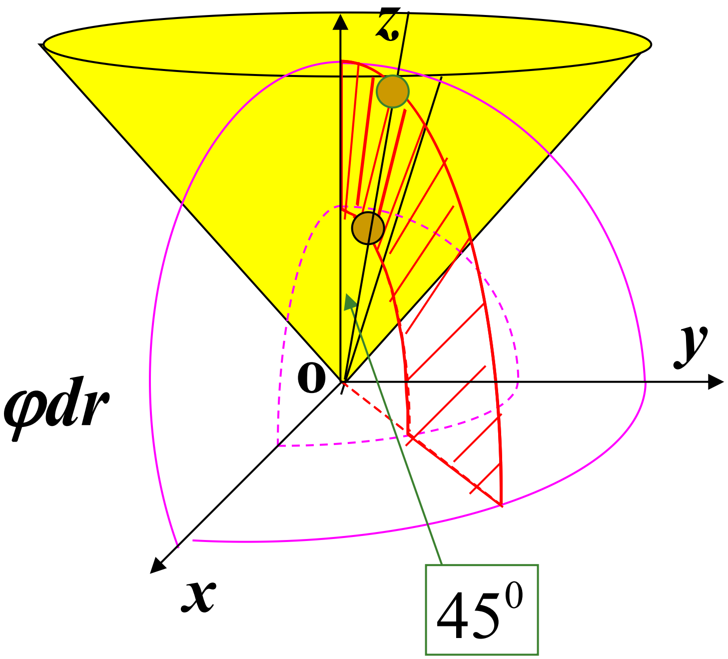
二、求由下列曲面所围 成的均匀密度体的重心

$$x^2+y^2+z^2 \ge 1 \quad x^2+y^2+z^2 \le 16 \quad z \ge \sqrt{x^2+y^2}$$

解 由对称性得，

$$\bar{x}=\bar{y}=0$$

$$M=\rho\int_0^{2\pi}d\theta\int_0^{\frac{\pi}{4}}d\varphi\int_1^4 r^2\sin\varphi dr$$



$$\bar{z}=\frac{1}{M}\rho\int_0^{2\pi}d\theta\int_0^{\frac{\pi}{4}}d\varphi\int_1^4 r^3\sin\varphi\cos\varphi dr$$

三、求：由$2z = x^2 + y^2$ ，$(x^2+y^2)^2 = x^2 - y^2$，$z = 0$所围立体体积

解 $V = \iint\limits_D dxdy\int_0^{\frac{x^2+y^2}{2}} dz \quad D:(x^2+y^2)^2 \le x^2 - y^2$

$$= 4\iint\limits_{D/4} dxdy\int_0^{\frac{x^2+y^2}{2}} dz = 4\iint\limits_{D/4} \frac{x^2+y^2}{2} dxdy$$

$$= 4\iint\limits_{D/4} \frac{\rho^2}{2}\rho d\rho d\theta = 4\int_0^{\frac{\pi}{4}} d\theta \int_0^{\sqrt{\cos 2\theta}} \frac{\rho^3}{2} d\rho$$

$$= 4\int_0^{\frac{\pi}{4}} \frac{1}{8}[\rho^4]_0^{\sqrt{\cos 2\theta}} d\theta = 4\int_0^{\frac{\pi}{4}} \frac{\cos^2 2\theta}{8} d\theta = \frac{\pi}{16}$$